Quantum Information and Computing - Lecture 2

Benjamin Bennetzen[1], Nikolaj Rossander Kristensen[1]

[1]Department of Computer Science, Aalborg University

September 17, 2025

AALBORG UNIVERSITY

Who are we? And context for the course

- Nikolaj Rossander Kristensen, PhD. Student. Studies quantum communication.
- Benjamin Bennetzen, PhD. Student. Studies quantitative reasoning for quantum programs.

Figure: Nikolaj

Figure: Benjamin

Who are we? And context for the course

- Nikolaj Rossander Kristensen, PhD. Student. Studies quantum communication.
- Benjamin Bennetzen, PhD. Student. Studies quantitative reasoning for quantum programs.

F24
First time the course was held, Benjamin and Nikolaj take the course.

E24

F25
Nikolaj teaches a guest lecture.

E25
Emergency takeover by Benjamin and Nikolaj.

F26
Nikolaj and Benjamin officially get to teach the course.

Figure: Nikolaj

Figure: Benjamin

What will be covered in the first part of the course

In this first part we will focus on the abstractions needed for programming quantum computers. We will touch on some of the physical aspects, but largely stay at an abstract level.

What will be covered in the first part of the course

In this first part we will focus on the abstractions needed for programming quantum computers. We will touch on some of the physical aspects, but largely stay at an abstract level.

This includes: Qubits, quantum gates, measurement, circuit model, superposition, entanglement, pure and mixed states, teleportation, and the basics quantum programming.

What will be covered in the first part of the course

In this first part we will focus on the abstractions needed for programming quantum computers. We will touch on some of the physical aspects, but largely stay at an abstract level.

This includes: Qubits, quantum gates, measurement, circuit model, superposition, entanglement, pure and mixed states, teleportation, and the basics quantum programming.

The second part of the course will have go into more depth about the physical aspects of quantum computers and will cover in detail some of the well known quantum algorithms.

What will be covered in the lecture

- Complex numbers
- Hilbert spaces
- Qubit
- Dirac notation
- States
- Measurement

Complex numbers

Definition (Complex number)

A complex number $z \in \mathbb{C}$ is a number that can be written in the form:

$$z = a + bi$$

Such that $a, b \in \mathbb{R}$ and i is the imaginary unit $i^2 = -1$.

Arithmetic for complex numbers

Definition (Addition of complex numbers)

$$(a + bi) + (c + di) = (a + c) + (b + d)i$$

Definition (Multiplication of complex numbers)

$$(a + bi)(c + di) = ac + adi + cbi + bdi^2 = ac + (ad + cb)i - bd$$

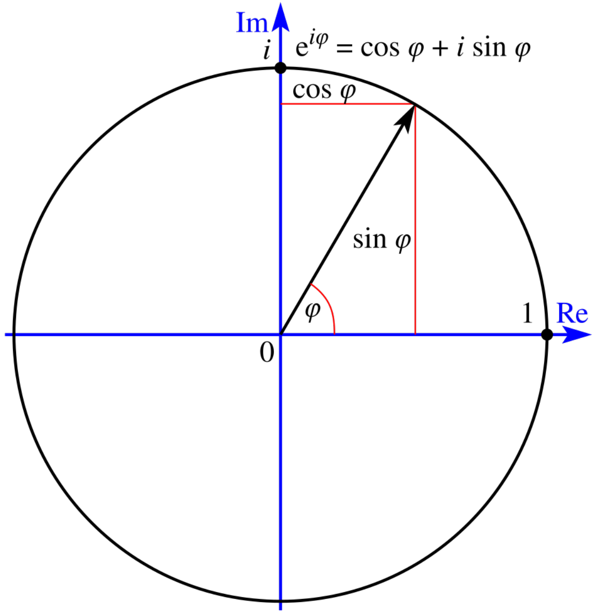
Geometric interpretation

One can interpret a complex number $a + bi$ as a vector $\begin{pmatrix} a \\ b \end{pmatrix}$.

- Addition corresponds to vector addition
- Multiplication corresponds to a scaling of the length and then a rotation.

The latter is more easily seen when you know Eulers formula $e^{xi} = \cos(x) + i\sin(x)$, meaning a complex number $re^{\theta i}$ has a length of r and angle θ. Multiplying two such numbers yields:

$$\left(re^{\theta i}\right)\left(r' e^{\theta' i}\right) = (rr')e^{(\theta+\theta')i}$$



Complex conjugate and absolute value

Definition (Complex conjugate)

The complex conjugate of $z = a + bi \in \mathbb{C}$ is:

$$z^* = a - bi$$

The notation $\overline{z}$ is also very common.

Geometric interpretation. A flip around the real axis.

Definition (Absolute value)

The absolute value of a complex number z is given by:

$$|z| = |a + bi| = \sqrt{a^2 + b^2}$$

Geometric interpretation. The length of the vector.

Bit physically

A classical bit is any physical system with two stable, distinguishable states.

Bit physically

A classical bit is any physical system with two stable, distinguishable states.

Examples:

- Voltage level in a wire: high / low
- Charge in a capacitor: present / absent
- Magnetization of a domain: up / down
- Hole in a punch card: yes / no

Bit physically

A classical bit is any physical system with two stable, distinguishable states.

Examples:

- Voltage level in a wire: high / low
- Charge in a capacitor: present / absent
- Magnetization of a domain: up / down
- Hole in a punch card: yes / no

The key property is stability: reading the bit does not change its value.

Qubit physically

A qubit is any object with quantum behavior, which has distinct states. Examples:

Qubit physically

A qubit is any object with quantum behavior, which has distinct states. Examples:

- Photon
  - Polarization: Horizontal, Vertical
- Atom
  - Energy: Excited, Ground
  - Spin: Up, Down

Qubit physically

A qubit is any object with quantum behavior, which has distinct states. Examples:

- Photon
  - Polarization: Horizontal, Vertical
- Atom
  - Energy: Excited, Ground
  - Spin: Up, Down

Superposition means that it is in a linear combination of basis states, however reading the state means that it collapses to one of them.

Qubit physically

A qubit is any object with quantum behavior, which has distinct states. Examples:

- Photon
  - Polarization: Horizontal, Vertical
- Atom
  - Energy: Excited, Ground
  - Spin: Up, Down

Superposition means that it is in a linear combination of basis states, however reading the state means that it collapses to one of them.

It does not really matter what underlying platform it is. The math describing a qubit is the same.

Qubit

Definition (Bit)

A bit is an element in the set $\{0, 1\}$.

Qubit

Definition (Bit)

A bit is an element in the set $\{0, 1\}$.

Definition (Qubit)

A qubit is a unit vector in the complex Hilbert space $\mathbb{C}^2$.

Qubit

Definition (Bit)

A bit is an element in the set $\{0, 1\}$.

Definition (Qubit)

A qubit is a unit vector in the complex Hilbert space $\mathbb{C}^2$.

What does this mean?

Qubit

Definition (Bit)

A bit is an element in the set $\{0, 1\}$.

Definition (Qubit)

A qubit is a unit vector in the complex Hilbert space $\mathbb{C}^2$.

What does this mean?

The definition takes a bit of unpacking, but afterwards you should hopefully see that a single qubit, is not as complex as the definition leads one to believe.

Vector Space

Definition (Vector space)

A vector space over a field F is an non-empty set V together with the operations $+ : V \times V \to V$ and $\cdot : F \times V \to V$. The latter usually denoted by juxtaposition of a scalar and a vector. Let $u, v, w \in V$ and $a, b \in F$ then the following axioms should be satisfied.

$$u + (v + w) = (u + v) + w \qquad a(bv) = (ab)v$$

$$u + v = v + u \qquad \exists 1 \in F . \forall v \in V . 1v = v$$

$$\exists 0 \in V . \forall v \in V . v + 0 = v \qquad a(u + v) = au + av$$

$$\forall v \in V . \exists -v \in V . v + (-v) = 0 \qquad (a + b)v = (av + bv)$$

Vector Space

Definition (Vector space)

A vector space over a field F is an non-empty set V together with the operations $+ : V \times V \to V$ and $\cdot : F \times V \to V$. The latter usually denoted by juxtaposition of a scalar and a vector. Let $u, v, w \in V$ and $a, b \in F$ then the following axioms should be satisfied.

$$
\begin{aligned}
u + (v + w) &= (u + v) + w & a(bv) &= (ab)v \\
u + v &= v + u & \exists 1 \in F . \forall v \in V . 1v &= v \\
\exists 0 \in V . \forall v \in V . v + 0 &= v & a(u + v) &= au + av \\
\forall v \in V . \exists -v \in V . v + (-v) &= 0 & (a + b)v &= (av + bv)
\end{aligned}
$$

Examples include the vector space $\mathbb{R}^n$ over $\mathbb{R}$ and $\mathbb{C}^n$ over $\mathbb{C}$, with the usual vector addition and scalar multiplication.

Inner product space

Definition (Inner product space)

An inner product space is a vector space V over a field $F \in \{\mathbb{R}, \mathbb{C}\}$ together with an inner product $\langle \cdot, \cdot \rangle : V \times V \to F$. The following should hold for all $u, v, w \in V$ and $a, b \in F$.

$$\langle u, v \rangle = \langle v, u \rangle^*$$
$$\langle au + bv, w \rangle = a\langle u, w \rangle + b\langle v, w \rangle$$
$$u \neq 0 \Rightarrow \langle u, u \rangle > 0$$

Note conjugate symmetry implies that $\langle u, u \rangle \in \mathbb{R}$.

Inner product space

Definition (Inner product space)

An inner product space is a vector space V over a field $F \in \{\mathbb{R}, \mathbb{C}\}$ together with an inner product $\langle \cdot, \cdot \rangle : V \times V \to F$. The following should hold for all $u, v, w \in V$ and $a, b \in F$.

$$\langle u, v \rangle = \langle v, u \rangle^*$$
$$\langle au + bv, w \rangle = a\langle u, w \rangle + b\langle v, w \rangle$$
$$u \neq 0 \Rightarrow \langle u, u \rangle > 0$$

Note conjugate symmetry implies that $\langle u, u \rangle \in \mathbb{R}$.

The previously mentioned vector spaces are also inner product spaces with the inner product being given by $\langle \binom{a}{b}, \binom{c}{d} \rangle = ac^* + bd^*$.

Metric space

Definition (Metric space)

A metric space is a pair (M, d) where M is a set and $d : M \times M \to \mathbb{R}$ is a metric, satisfying the following for all $x, y, z \in M$:

$$d(x,x) = 0$$

$$x \neq y \Rightarrow d(x,y) > 0$$

$$d(x,y) = d(y,x)$$

$$d(x,z) \leq d(x,y) + d(y,z)$$

Metric space

Definition (Metric space)

A metric space is a pair (M, d) where M is a set and $d : M \times M \to \mathbb{R}$ is a metric, satisfying the following for all $x, y, z \in M$:

$$d(x,x) = 0$$
$$x \neq y \Rightarrow d(x,y) > 0$$
$$d(x,y) = d(y,x)$$
$$d(x,z) \leq d(x,y) + d(y,z)$$

$\mathbb{R}^2$ endowed with Euclidian distance (how the crow flies) or the Manhattan distance (zig-zag), are both examples of metric spaces.

Metric space

Definition (Metric space)

A metric space is a pair (M, d) where M is a set and $d : M \times M \to \mathbb{R}$ is a metric, satisfying the following for all $x, y, z \in M$:

$$d(x, x) = 0$$

$$x \neq y \Rightarrow d(x, y) > 0$$

$$d(x, y) = d(y, x)$$

$$d(x, z) \leq d(x, y) + d(y, z)$$

$\mathbb{R}^2$ endowed with Euclidian distance (how the crow flies) or the Manhattan distance (zig-zag), are both examples of metric spaces.

A metric space is called complete if every Cauchy sequence of points in M has a limit that is also in M. (do not worry about this)

Hilbert space

Definition (Hilbert space)

A Hilbert space is a real or complex inner product space that is also a complete metric space with respect to metric induced by the inner product.

$$\|u\| = \sqrt{\langle u, u \rangle}$$
$$d(u, v) = \|u - v\|$$

Qubit... again

Definition (Qubit)

A qubit is a unit vector in the complex Hilbert space $\mathbb{C}^2$.

With a better understanding of the terminology in this definition, we can now state what a qubit is in a way that is easier to grasp.

Qubit... again

Definition (Qubit)

A qubit is a unit vector in the complex Hilbert space $\mathbb{C}^2$.

With a better understanding of the terminology in this definition, we can now state what a qubit is in a way that is easier to grasp.

A qubit is a vector $\binom{a}{b}$ with $a, b \in \mathbb{C}$ s.t $\|\binom{a}{b}\| = 1$.

$$\left\|\begin{pmatrix} a \\ b \end{pmatrix}\right\| = \sqrt{\left\langle \begin{pmatrix} a \\ b \end{pmatrix}, \begin{pmatrix} a \\ b \end{pmatrix} \right\rangle} = \sqrt{aa^* + bb^*} = \sqrt{|a|^2 + |b|^2} \iff |a|^2 + |b|^2 = 1$$

Qubit... again

Definition (Qubit)

A qubit is a unit vector in the complex Hilbert space $\mathbb{C}^2$.

With a better understanding of the terminology in this definition, we can now state what a qubit is in a way that is easier to grasp.

A qubit is a vector $\begin{pmatrix} a \\ b \end{pmatrix}$ with $a, b \in \mathbb{C}$ s.t $\| \begin{pmatrix} a \\ b \end{pmatrix} \| = 1$.

$$\| \begin{pmatrix} a \\ b \end{pmatrix} \| = \sqrt{\langle \begin{pmatrix} a \\ b \end{pmatrix}, \begin{pmatrix} a \\ b \end{pmatrix} \rangle} = \sqrt{aa^* + bb^*} = \sqrt{|a|^2 + |b|^2} \Longleftrightarrow |a|^2 + |b|^2 = 1$$

Note states that differ by a global phase represent the same physical state.

$$|\varphi\rangle \equiv e^{ix} \; |\varphi\rangle$$

Bra-Ket / Dirac notation

It is cumbersome to write vectors all the time. Let us denote the vectors $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$ by the kets $|0\rangle$ and $|1\rangle$ respectively.

$\{|0\rangle, |1\rangle\}$ is called the computational basis, and it spans all of $\mathbb{C}^2$, meaning we can write arbitrary qubit states as a linear combination of $|0\rangle$ and $|1\rangle$:

$$|\varphi\rangle = a|0\rangle + b|1\rangle \text{ where } |a|^2 + |b|^2 = 1$$

Bra-Ket / Dirac notation

It is cumbersome to write vectors all the time. Let us denote the vectors $\begin{pmatrix}1\\0\end{pmatrix}$ and $\begin{pmatrix}0\\1\end{pmatrix}$ by the kets $|0\rangle$ and $|1\rangle$ respectively.

$\{|0\rangle, |1\rangle\}$ is called the computational basis, and it spans all of $\mathbb{C}^2$, meaning we can write arbitrary qubit states as a linear combination of $|0\rangle$ and $|1\rangle$:

$$|\varphi\rangle = a|0\rangle + b|1\rangle \text{ where } |a|^2 + |b|^2 = 1$$

$|\varphi\rangle$ is called a ket. We then define a bra $\langle\varphi|$ as the conjugate transpose of a ket.

$$\left(|\varphi\rangle^T\right)^* = |\varphi\rangle^\dagger = \langle\varphi|$$

Bra-Ket / Dirac notation

It is cumbersome to write vectors all the time. Let us denote the vectors $\begin{pmatrix}1\\0\end{pmatrix}$ and $\begin{pmatrix}0\\1\end{pmatrix}$ by the kets $|0\rangle$ and $|1\rangle$ respectively.

$\{|0\rangle, |1\rangle\}$ is called the computational basis, and it spans all of $\mathbb{C}^2$, meaning we can write arbitrary qubit states as a linear combination of $|0\rangle$ and $|1\rangle$:

$$|\varphi\rangle = a|0\rangle + b|1\rangle \text{ where } |a|^2 + |b|^2 = 1$$

$|\varphi\rangle$ is called a ket. We then define a bra $\langle\varphi|$ as the conjugate transpose of a ket.

$$\left(|\varphi\rangle^T\right)^* = |\varphi\rangle^\dagger = \langle\varphi|$$

We can also form a bra-ket!

$$\langle\varphi|\psi\rangle = \langle\varphi, \psi\rangle$$

Bra-Ket / Dirac notation

It is cumbersome to write vectors all the time. Let us denote the vectors $\begin{pmatrix}1\\0\end{pmatrix}$ and $\begin{pmatrix}0\\1\end{pmatrix}$ by the kets $|0\rangle$ and $|1\rangle$ respectively.

$\{|0\rangle, |1\rangle\}$ is called the computational basis, and it spans all of $\mathbb{C}^2$, meaning we can write arbitrary qubit states as a linear combination of $|0\rangle$ and $|1\rangle$:

$$|\varphi\rangle = a|0\rangle + b|1\rangle \text{ where } |a|^2 + |b|^2 = 1$$

$|\varphi\rangle$ is called a ket. We then define a bra $\langle\varphi|$ as the conjugate transpose of a ket.

$$\left(|\varphi\rangle^T\right)^* = |\varphi\rangle^\dagger = \langle\varphi|$$

We can also form a bra-ket!

$$\langle\varphi|\psi\rangle = \langle\varphi, \psi\rangle$$

Throughout the course you will hopefully see why Dirac notation is so convenient.

Bloch Sphere interpretation

The Bloch sphere is a useful visual representation of a qubit, that quickly gives one a visual intuition for many concepts. However, the analogy quickly falls apart in multi qubit systems.

Bloch Sphere interpretation

The Bloch sphere is a useful visual representation of a qubit, that quickly gives one a visual intuition for many concepts. However, the analogy quickly falls apart in multi qubit systems.

$$|\psi\rangle = a|0\rangle + b|1\rangle = r_a e^{i\theta_a}|0\rangle + r_b e^{i\theta_b}|1\rangle \equiv$$
$$e^{-i\theta_a}\left(r_a e^{i\theta_a}|0\rangle + r_b e^{i\theta_b}|1\rangle\right) = r_a\ |0\rangle + r_b e^{i(\theta_b-\theta_a)}|1\rangle$$

Bloch Sphere interpretation

The Bloch sphere is a useful visual representation of a qubit, that quickly gives one a visual intuition for many concepts. However, the analogy quickly falls apart in multi qubit systems.

$$|\psi\rangle = a|0\rangle + b|1\rangle = r_a e^{i\theta_a}|0\rangle + r_b e^{i\theta_b}|1\rangle \equiv$$
$$e^{-i\theta_a}\left(r_a e^{i\theta_a}|0\rangle + r_b e^{i\theta_b}|1\rangle\right) = r_a \ |0\rangle + r_b e^{i(\theta_b - \theta_a)}|1\rangle$$

For some x and y we have $r_b e^{i(\theta_b - \theta_a)} = x + yi$, and for convenience let us denote r_b by z. Due to normalization we know that:

$$|r_a|^2 + |r_b|^2 = z^2 + (x + yi)(x - yi) = z^2 + x^2 + y^2 = 1$$

Bloch Sphere interpretation

The Bloch sphere is a useful visual representation of a qubit, that quickly gives one a visual intuition for many concepts. However, the analogy quickly falls apart in multi qubit systems.

$$|\psi\rangle = a|0\rangle + b|1\rangle = r_a e^{i\theta_a}|0\rangle + r_b e^{i\theta_b}|1\rangle \equiv$$
$$e^{-i\theta_a}\left(r_a e^{i\theta_a}|0\rangle + r_b e^{i\theta_b}|1\rangle\right) = r_a \ |0\rangle + r_b e^{i(\theta_b - \theta_a)}|1\rangle$$

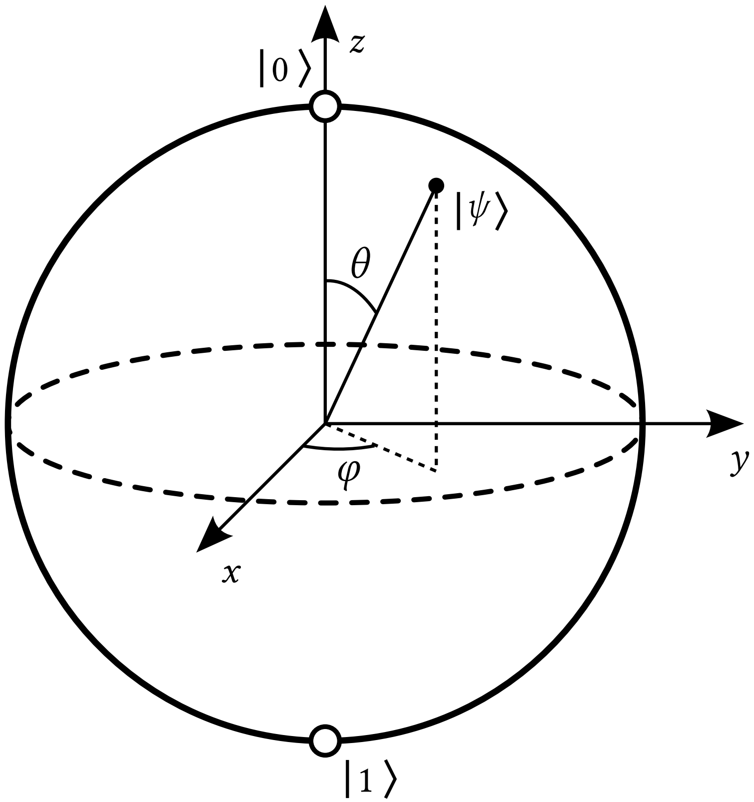
For some x and y we have $r_b e^{i(\theta_b - \theta_a)} = x + yi$, and for convenience let us denote r_b by z. Due to normalization we know that:

$$|r_a|^2 + |r_b|^2 = z^2 + (x + yi)(x - yi) = z^2 + x^2 + y^2 = 1$$

This can be mapped to spherical coordinates. $x = \sin(\theta)\cos(\varphi)$, $y = \sin(\theta)\sin(\varphi)$ and $z = \cos(\theta)$. Thus:

$$|\psi\rangle = z|0\rangle + (x + yi)|1\rangle = \cos(\theta)|0\rangle + (\sin(\theta)\cos(\varphi) + \sin(\theta)\sin(\varphi)i)|1\rangle$$

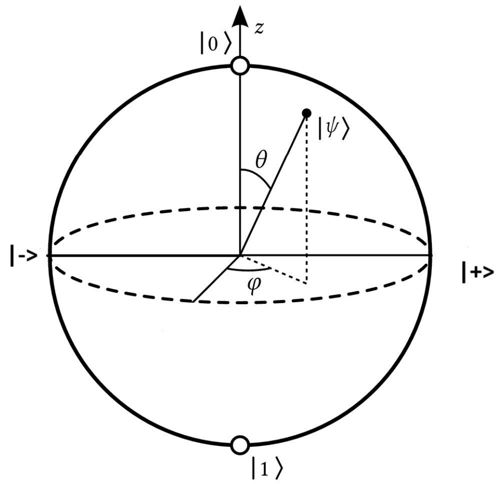
Bloch sphere



Different bases

As mentioned the set $\{|0\rangle, |1\rangle\}$ is the computational basis. In reality there are infinitely many such bases, pick two points on opposite ends of the Bloch sphere and you have one. Mathematically this corresponds to the vectors being orthonormal. $\| \, |\varphi\rangle \, \| = \| \, |\psi\rangle \, \| = 1$ and $\langle \varphi | \psi \rangle = 0$.

Another common basis is the Hadamard basis, written as $| +\rangle$ and $| -\rangle$.



Superposition

A superposition of a qubit is basis dependent.

We can write the states states $|+\rangle$ and $|-\rangle$ as linear combinations of the computational basis.

$$|+\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle \quad |-\rangle = \frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}|1\rangle$$

From the perspective of the computational basis, the states $|+\rangle$ and $|-\rangle$ are in a superposition between $|0\rangle$ and $|1\rangle$. However, if we write them in the hadamard basis then it will become clear that they are not in a superposition between $|+\rangle$ and $|-\rangle$.

$$|+\rangle = 1|+\rangle + 0|-\rangle \quad |-\rangle = 0|+\rangle + 1|-\rangle$$

Superposition

A superposition of a qubit is basis dependent.

We can write the states states $|+\rangle$ and $|-\rangle$ as linear combinations of the computational basis.

$$|+\rangle = \frac{1}{\sqrt{2}}\,|0\rangle + \frac{1}{\sqrt{2}}\,|1\rangle \quad |-\rangle = \frac{1}{\sqrt{2}}\,|0\rangle - \frac{1}{\sqrt{2}}\,|1\rangle$$

From the perspective of the computational basis, the states $|+\rangle$ and $|-\rangle$ are in a superposition between $|0\rangle$ and $|1\rangle$. However, if we write them in the hadamard basis then it will become clear that they are not in a superposition between $|+\rangle$ and $|-\rangle$.

$$|+\rangle = 1|+\rangle + 0|-\rangle \quad |-\rangle = 0|+\rangle + 1|-\rangle$$

Next we will talk about measurements, hopefully this should make more sense then.

Measurements

In quantum computation we cannot simply **read** the data stored in a qubit. Unlike classical bits, observing a qubit fundamentally changes its state.

A qubit can be in a superposition of some arbitrary basis $\{|\varphi\rangle, |\varphi^{\perp}\rangle\}$

$$a|\varphi\rangle + b|\varphi^{\perp}\rangle$$

but measurement forces it to choose a definite outcome, we say that the states collapses.

- It collapses to either $|\varphi\rangle$ or $|\varphi^{\perp}\rangle$.
- The outcome is determined by the amplitudes a and b.

Measurements

In quantum computation we cannot simply **read** the data stored in a qubit. Unlike classical bits, observing a qubit fundamentally changes its state.

A qubit can be in a superposition of some arbitrary basis $\{|\varphi\rangle, |\varphi^\perp\rangle\}$

$$a|\varphi\rangle + b|\varphi^\perp\rangle$$

but measurement forces it to choose a definite outcome, we say that the states collapses.

- It collapses to either $|\varphi\rangle$ or $|\varphi^\perp\rangle$.
- The outcome is determined by the amplitudes a and b.
- $|a|^2 =$ probability the qubit collapses to $|\varphi\rangle$, and $|b|^2 =$ probability the qubit collapses to $|\varphi^\perp\rangle$. Notice we have $|a|^2 + |b|^2 = 1$.

Measurements

In quantum computation we cannot simply **read** the data stored in a qubit. Unlike classical bits, observing a qubit fundamentally changes its state.

A qubit can be in a superposition of some arbitrary basis $\{|\varphi\rangle, |\varphi^\perp\rangle\}$

$$a|\varphi\rangle + b|\varphi^\perp\rangle$$

but measurement forces it to choose a definite outcome, we say that the states collapses.

- It collapses to either $|\varphi\rangle$ or $|\varphi^\perp\rangle$.
- The outcome is determined by the amplitudes a and b.
- $|a|^2 =$ probability the qubit collapses to $|\varphi\rangle$, and $|b|^2 =$ probability the qubit collapses to $|\varphi^\perp\rangle$. Notice we have $|a|^2 + |b|^2 = 1$.
- Measurements are a little more complex, but know that mathematically we can turn any basis for a Hilbert space into a type of measurement.

Example

The state $|+\rangle$;

$$|+\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$$

Will when measured in the computational basis with 50% chance become $|0\rangle$ and with 50% chance become $|1\rangle$. Since:

$$|\frac{1}{\sqrt{2}}|^2 = \frac{1}{2}$$

Example

The state $|+\rangle$;

$$|+\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$$

Will when measured in the computational basis with 50% chance become $|0\rangle$ and with 50% chance become $|1\rangle$. Since:

$$|\frac{1}{\sqrt{2}}|^2 = \frac{1}{2}$$

Suppose we measure in the hadamard basis.

$$|+\rangle = 1|+\rangle + 0|-\rangle$$

It will then collapse to $|+\rangle$ deterministically, because $1^2 = 1$.

Example

The state $|+\rangle$;

$$|+\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$$

Will when measured in the computational basis with 50% chance become $|0\rangle$ and with 50% chance become $|1\rangle$. Since:

$$|\frac{1}{\sqrt{2}}|^2 = \frac{1}{2}$$

Suppose we measure in the hadamard basis.

$$|+\rangle = 1|+\rangle + 0|-\rangle$$

It will then collapse to $|+\rangle$ deterministically, because $1^2 = 1$.

Next we give a more mathematically rigorous explanation of measurements.

Outer product

Definition (Outer product)

Let $|\varphi\rangle$ and $|\psi\rangle$ be kets in $\mathbb{C}^n$.

The outer product is the linear operator:

$$|\varphi\rangle\langle\psi|$$

defined by its action on a state $|x\rangle$:

$$(|\varphi\rangle\langle\psi|)\ |x\rangle = \langle\psi|x\rangle\ |\varphi\rangle$$

Outer product

More concretely, if:

$$|\varphi\rangle = \begin{pmatrix} \varphi_1 \\ \varphi_2 \\ \cdots \\ \varphi_n \end{pmatrix} \quad \text{and} \quad |\psi\rangle = \begin{pmatrix} \psi_1 \\ \psi_2 \\ \cdots \\ \psi_n \end{pmatrix}$$

Then the outer product has matrix representation:

$$|\varphi\rangle\langle\psi| = \begin{pmatrix} \varphi_1\psi_1^* & \varphi_1\psi_2^* & \cdots & \varphi_1\psi_n^* \\ \varphi_2\psi_1^* & \varphi_2\psi_2^* & \cdots & \varphi_2\psi_n^* \\ \vdots & \vdots & \ddots & \vdots \\ \varphi_n\psi_1^* & \varphi_n\psi_2^* & \cdots & \varphi_n\psi_n^* \end{pmatrix}$$

Projectors

Definition (Linear operator)

Let U and V be vector spaces, then a mapping $A : U \to V$ is linear if:

$$A(ax + by) = aAx + bAy$$

Definition (Projector)

A projector P is a linear operator such that:

$$P^2 = P \quad \text{and} \quad P^\dagger = P$$

Projectors

Definition (Linear operator)

Let U and V be vector spaces, then a mapping $A : U \to V$ is linear if:

$$A(ax + by) = aAx + bAy$$

Definition (Projector)

A projector P is a linear operator such that:

$$P^2 = P \quad \text{and} \quad P^\dagger = P$$

The operators

$$|0\rangle\langle 0| \quad \text{and} \quad |1\rangle\langle 1|$$

are projectors.

Measurement rule

Measuring in the computational basis corresponds to:

$$P_0 = |0\rangle\langle 0| \quad P_1 = |1\rangle\langle 1|$$

Measurement rule

Measuring in the computational basis corresponds to:

$$P_0 = |0\rangle\langle 0| \quad P_1 = |1\rangle\langle 1|$$

Given a state $|\psi\rangle$, the probability of outcome i is:

$$p(i) = \langle\psi|\, P_i\, |\psi\rangle$$

Measurement rule

Measuring in the computational basis corresponds to:

$$P_0 = |0\rangle\langle 0| \quad P_1 = |1\rangle\langle 1|$$

Given a state $|\psi\rangle$, the probability of outcome i is:

$$p(i) = \langle\psi|\ P_i\ |\psi\rangle$$

If outcome i is observed, the post-measurement state is:

$$|\psi_i\rangle = \frac{P_i\ |\psi\rangle}{\sqrt{\langle\psi|P_i|\psi\rangle}}$$

POVM

Definition (Positive operator valued measure (POVM))

A POVM is a set of operators positive semi-definite self-adjoint operators $\{E_i\}_{i\in\{1,\ldots,n\}}$ such that:

$$\sum_i E_i = I$$

Projectors are a special case $E_i = P_i$.

POVM

Definition (Positive operator valued measure (POVM))

A POVM is a set of operators positive semi-definite self-adjoint operators $\{E_i\}_{i\in\{1,\dots,n\}}$ such that:

$$\sum_i E_i = I$$

Projectors are a special case $E_i = P_i$.

Do not worry about the jargon. Just know that given an orthonormal basis $\{|\varphi_i\rangle\}_{i,\dots,n}$, we can define:

$$E_i = |\varphi_i\rangle\langle\varphi_i|$$

Then $\{E_i\}_i$ is a POVM.

Example: $|+\rangle$ measured in computational basis (POVM)

Consider the state:

$$|+\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$$

Example: $|+\rangle$ measured in computational basis (POVM)

Consider the state:

$$|+\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$$

The computational basis defines the POVM:

$$E_0 = |0\rangle\langle 0| \quad E_1 = |1\rangle\langle 1|$$

Example: $|+\rangle$ measured in computational basis (POVM)

Consider the state:

$$|+\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$$

The computational basis defines the POVM:

$$E_0 = |0\rangle\langle 0| \quad E_1 = |1\rangle\langle 1|$$

The probability of outcome i is:

$$p(i) = \langle +| \, E_i \, |+\rangle$$

Example: $|+\rangle$ measured in computational basis (POVM)

Consider the state:

$$|+\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$$

The computational basis defines the POVM:

$$E_0 = |0\rangle\langle 0| \quad E_1 = |1\rangle\langle 1|$$

The probability of outcome i is:

$$p(i) = \langle + | \, E_i \, | + \rangle$$

$$p(0) = \langle + | 0\rangle\langle 0 | + \rangle = |\frac{1}{\sqrt{2}}|^2 = \frac{1}{2}$$

$$p(1) = \langle + | 1\rangle\langle 1 | + \rangle = |\frac{1}{\sqrt{2}}|^2 = \frac{1}{2}$$

Example: $|+\rangle$ measured in computational basis (POVM)

Consider the state:

$$|+\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$$

The computational basis defines the POVM:

$$E_0 = |0\rangle\langle 0| \quad E_1 = |1\rangle\langle 1|$$

The probability of outcome i is:

$$p(i) = \langle +| \; E_i \; |+\rangle$$

$$p(0) = \langle +|0\rangle\langle 0|+\rangle = |\frac{1}{\sqrt{2}}|^2 = \frac{1}{2}$$

$$p(1) = \langle +|1\rangle\langle 1|+\rangle = |\frac{1}{\sqrt{2}}|^2 = \frac{1}{2}$$

If outcome 0 is observed, the state collapses to $|0\rangle$. If outcome 1 is observed, the state collapses to $|1\rangle$.

Example: $|+\rangle$ measured in Hadamard basis (POVM)

The Hadamard basis defines the POVM:

$$E_+ = |+\rangle\langle+| \quad E_- = |-\rangle\langle-|$$

Example: $|+\rangle$ measured in Hadamard basis (POVM)

The Hadamard basis defines the POVM:

$$E_+ = |+\rangle\langle+| \quad E_- = |-\rangle\langle-|$$

Again, the probability of outcome i is:

$$p(i) = \langle+| E_i |+\rangle$$

Example: $|+\rangle$ measured in Hadamard basis (POVM)

The Hadamard basis defines the POVM:

$$E_+ = |+\rangle\langle+| \quad E_- = |-\rangle\langle-|$$

Again, the probability of outcome i is:

$$p(i) = \langle+| \, E_i \, |+\rangle$$

$$p(+) = \langle+|+\rangle^2 = 1$$

$$p(-) = \langle-|+\rangle^2 = 0$$

Example: $|+\rangle$ measured in Hadamard basis (POVM)

The Hadamard basis defines the POVM:

$$E_+ = |+\rangle\langle+| \quad E_- = |-\rangle\langle-|$$

Again, the probability of outcome i is:

$$p(i) = \langle+| \, E_i \, |+\rangle$$

$$p(+) = \langle+|+\rangle^2 = 1$$

$$p(-) = \langle-|+\rangle^2 = 0$$

The measurement outcome is deterministic, and the post-measurement state is $|+\rangle$.

Exercises

Complex numbers

Prove that the following properties holds for $z \in \mathbb{C}$:

1. $(z^*)^* = z$
2. $(z_1 z_2)^* = z_1^* z_2^*$
3. $(z_1 + z_2)^* = z_1^* + z_2^*$
4. $|z|^2 = z^* z$

Superposition

1. Write $|0\rangle$ and $|1\rangle$ as a superposition between $|+\rangle$ and $|-\rangle$.

Measurement

1. Check that $|0\rangle\langle 0|$ and $|1\rangle\langle 1|$ are projectors.
2. In the last two examples we skipped the calculations of the state after measurement. Can you use the formula for the post measurement state to find them?
3. What happens if we measure the state $|+\rangle$ in the computational basis, and then the hadamard basis? What is if we did the other way around?